

AFL Chat + Prediction Assistant

Executive Report · Week 6 Day 5 Capstone

Product Goal

Deliver a domain-locked conversational assistant for AFL fans and analysts that answers factual questions about historical stats, provides head-to-head records, and generates calibrated match and player predictions — while refusing, by construction, to go outside AFL scope. The system is suitable for embedding in a sports media property (e.g. Web3Geeks) as a self-service stats and prediction tool, reducing manual research time for editors and giving readers an interactive alternative to static stat pages.

Architecture

The system is a LangGraph state machine with explicit routing — not a free-roaming agent. Every user turn flows through the same fixed pipeline:

Node	Responsibility
router.py	Regex classifier → intent (retrieval / prediction / factual / off_topic) + sub-type + raw entity extraction
resolver.py	Maps raw team/player/year mentions to dataset values; pronoun + year carryover across turns
retrieval_node.py	Runs the correct tool (head-to-head, season stats, game stats, team leaders) against real CSVs/parquet
prediction_node.py	Calls the trained Logistic Regression pipeline (match winner) or GBM (top player fantasy points)
validation_node.py	Checks tool output for errors; routes to clarify / fallback if needed
format_response_node.py	Single convergence point — EVERY prediction exits here with the 'not a certainty' disclaimer baked in
hardening.py	safe_node wraps every node; run_with_timeout caps slow tool calls; jailbreak/scope patterns block off-topic
api.py	FastAPI wrapper — session persistence, structured JSON logging, /chat + /health endpoints

The key architectural decision is that the disclaimer and probability framing are structural guarantees (code paths), not prompt instructions. There is no execution path through the graph that produces a prediction without them.

Evaluation Results

Category	Pass	Total	Pass rate
A: Factual / Retrieval Accuracy	7	7	100 %
B: Prediction Sanity	7	7	100 %
C: Scope Guardrails	6	7	86 %
D: Multi-turn Coherence	6	7	86 %
Overall	26	28	93 %

Categories A and B are 100 %. The single C failure (C7: generic tech question routed to factual instead of off_topic) is a known, narrow gap in the generic off-topic keyword list — a follow-up patch to _GENERIC_OFFTOPIC in router.py closes it. The single D failure (D3: 'instead' team-swap follow-up) is a genuine ambiguity (the system correctly asks for clarification rather than guessing) and is documented as a known hard case.

Model Benchmark

Model	Accuracy	F1	ROC-AUC
Logistic Regression (shipped)	64.47 %	63.58 %	71.80 %
Gradient Boosting (evaluated, not shipped)	63.31 %	61.94 %	72.14 %
Baseline: always higher-ladder team wins	63.19 %	62.02 %	63.20 %
Baseline: always home team wins	56.71 %	56.51 %	56.71 %

The shipped model beats the naive ladder-position baseline by 1.3 pp accuracy but by 8.6 pp ROC-AUC — meaning its probability calibration is meaningfully better even though its win/loss call accuracy is close. The honest framing: ladder position is already a strong predictor in AFL, so the model's value is in probability quality (useful for the 'not a certainty' framing the system is built around), not a dramatic accuracy jump.

Known Limitations

Data recency: Stats and ladder state reflect the last data load. Without a live data feed, predictions degrade as the season progresses past the last ingestion date.

Model accuracy ceiling: 64.5 % accuracy is real predictive signal above the 63.2 % naive baseline, but it is not a strong enough signal to use for betting or fantasy without further feature engineering.

Guardrail edge cases: C7 (generic tech questions) and any novel jailbreak phrasings not yet in the pattern list can slip through to the factual catch-all. The monitoring plan tracks off-topic leak rate to catch new patterns.

Multi-turn ambiguity: The 'instead' team-swap follow-up (D3) is not handled — the system asks for clarification, which is correct but adds friction in a natural conversation flow.

No live fixture list: Predictions for 'this week' are based on recent form, not a confirmed scheduled matchup, since no fixture API is wired in.

Recommended Next Steps

- Wire a live AFL data feed (Champion Data or AFL official API) to eliminate data recency lag.
- Add _GENERIC_OFFTOPIC patterns for 'how does X work' / 'explain X' where X is not AFL-related (closes C7).
- Implement LLM-based intent resolution for ambiguous follow-ups like D3, using the existing router.py gateway hook.
- Move session store from in-memory dict to Redis for multi-worker / multi-process API deployment.
- Add a cost-per-turn metric to the monitoring log once the router is upgraded to an LLM gateway call.
- Embed the chat UI in the target property's CMS and run a 2-week user acceptance test before full launch.